11	The International Space Station (ISS) orbits the Earth with uniform speed, above the Earth's atmosphere, at a constant height 420 km above the Earth's surface. What statement about the ISS is correct?	
	A	There is no gravitational force on the ISS.
	B	The resultant force on the ISS is in the direction it is moving.
	C	The ISS is not accelerating.
	D	The acceleration of the ISS is less than 9.81 m s^{-2}.
	Answer: D The gravitational force provides the centripetal force for the ISS to move in circular motion and the direction of the centripetal force is towards the centre of the Earth. Option A and B are incorrect. The acceleration is equal to the magnitude of the gravitational field strength $\left(\frac{GM_{Earth}}{r^2}\right)$. Since the acceleration on the Earth's surface is 9.81 m s^{-1} and the ISS is above the Earth's surface, its acceleration will be less than 9.81 m s^{-1}. Option D.	

12	A weather balloon, initially stationary on the ground, contains an ideal gas with an internal